

DWEP+

Revolutionizing Urban-Industrial Water Circularity:
A Sustainable Water Exchange Model for
Durgapur

The Durgapur Water Crisis

"The Damodar River basin currently exhibits critical thresholds of industrial and urban contamination, creating an immediate requirement for closed-loop computational resource recovery."

Durgapur's industrial giants face a severe water deficit while 70-80% of municipal wastewater is discharged untreated into the Damodar River.

 **Critical Scarcity:** Industrial units compete with residents for groundwater.

 **Heavy Metal Load:** River pollution costs the region ₹272 million annually in damage.

Ex.- Lead, Cadmium, Arsenic, Ammonia, Cyanide
Phenol etc.



Introducing DWEP+



Digital Water Exchange

An automated marketplace linking Municipal STPs with Industrial Demand Centers (Steel & Power Plants).



Zero Liquid Discharge

Enabling industries to achieve ZLD compliance by utilizing 10-40% treated municipal wastewater.

The Technical Edge



IoT Monitoring

Real-time sensing of pH, TDS, and Heavy Metals (Fe, Pb, Mn) to ensure industrial-grade quality control.



Blockchain Ledger

Immutable "Smart Contracts" to resolve legal liability and transparency at the point of water transfer.



AI Matching

Dynamic routing and pricing based on real-time industrial demand and municipal supply peaks.

Economic Feasibility

₹272M

40%

₹150C

Environmental Cost Savings

Reduction in Industrial Cost

Annual Revenue Potential

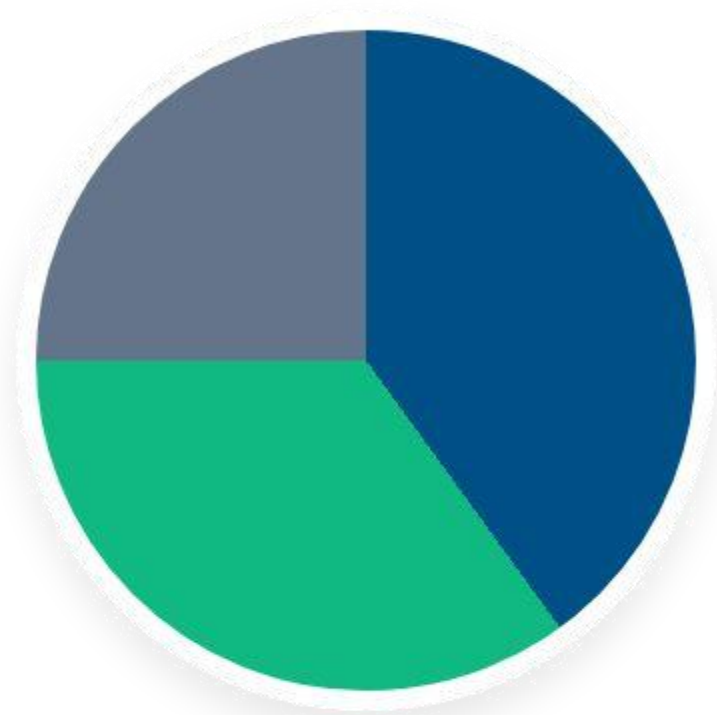
Metric	Value	Justification
Unit Revenue	₹45 / KL	40% cheaper than industrial freshwater
Unit Cost	₹25 / KL	Includes STP OpEx + Digital Layer
Unit Profit	₹20 / KL	44% Net Profit Margin
Market Size	200 MLD	Total Durgapur Municipal Output
Target Revenue	₹150 Cr/Year	Based on 50% market capture (105 MLD)

"Moving from Wastewater Disposal to Resource Recovery (Asset)."

Target Market Segments

Stakeholder	Pain Point	DWEP+ Value Proposition
Durgapur Steel Plant (DSP)	Specific Water Consumption high	40% Reuse potential for cooling towers
Thermal Power Stations	Groundwater extraction limits	Reliable supply of non-potable water
Municipal Corporation	Cost of STP maintenance	Revenue generation through water sales
Small-Scale Industries	Tanker water price volatility	Standardized pricing and digital tracking

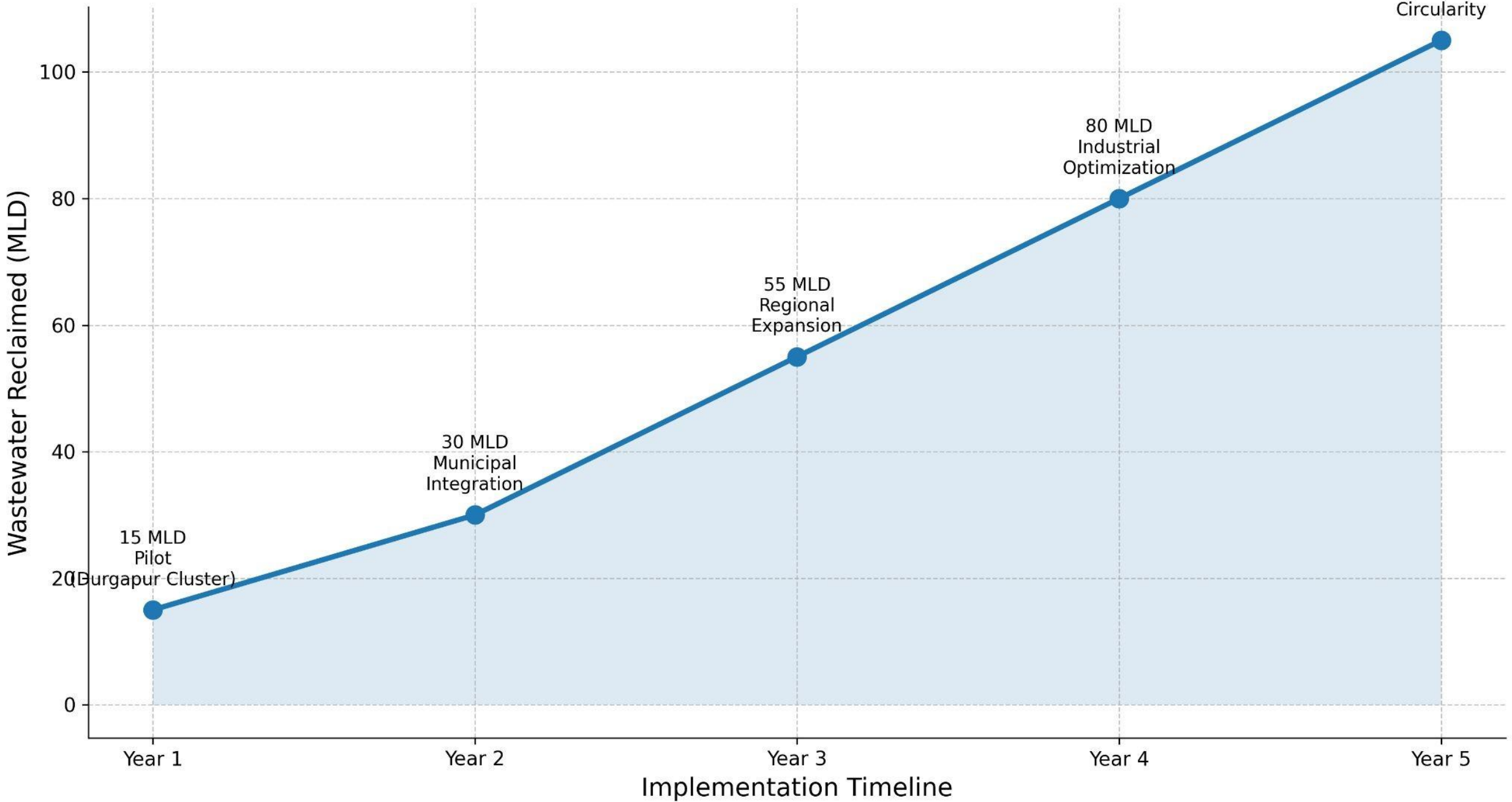
Sustainability Analysis



- Economic: Sustainable Revenue Model (40%)
- Environmental: Damodar River Restoration (35%)
- Social: Improved Public Health (25%)

DWEP+ fulfills the **Triple Bottom Line** by ensuring environmental engineering is economically self-sufficient.

Projected Wastewater Reclamation Growth



Policy Pivot

West Bengal Water Reuse Policy 2026

Aligning with the draft state mandate for industries to utilize 10-40% municipal wastewater.

DWEP+ provides the necessary digital infrastructure for the state to enforce this mandate while ensuring industrial competitiveness.



Implementation Roadmap



Beyond Durgapur



Asansol Hub

Replicating the model in the coal-rich industrial belt of Asansol.



Smart City Integration

Plugging into National Smart City missions for digital water management.



Haldia Port Cluster

Applying the exchange model to chemical and refining industrial zones.

Questions?

Thank you for your attention.

Team DeepAxe | NIT Durgapur Ideathon 2026

Building a Water-Secure Future, One Drop at a Time.